PRECISION IMU CARRIER

ARK ADIS16507

Precision 6-DoF MEMS IMU carrier based on the ADIS16507-2 or ADIS16507-3

The USA-built ARK ADIS16507 is a compact carrier for the Analog Devices ADIS16507 precision MEMS IMU, sold with the ADIS16507-2 or ADIS16507-3. It provides a factory-calibrated triaxial gyroscope (±500 °/s or ±2000 °/s) and triaxial accelerometer (±40 g) over SPI, with onboard 3.3 V regulation from a 5 V supply.

The 11-pin JST-GH connector matches the ARK PAB SPI pinout for direct use with ARK flight controllers and carriers.



KEY FEATURES

- ADIS16507-2 or ADIS16507-3 precision MEMS IMU
- Gyro: ±500 °/s (-2) or ±2000 °/s (-3)
- Accel: ±40 g
- Factory calibrated sensitivity, bias, and alignment
- SPI interface with data-ready
- Onboard 3.3 V LDO from 5 V input
- 11-pin JST-GH matching ARK PAB SPI pinout
- Includes cable

SPECIFICATIONS

ELECTRICAL	
Supply voltage	5 V
Supply current	41 mA average
IMU supply	3.3 V (onboard LDO)
Interface	SPI

ENVIRONMENTAL	
Operating temperature	−25 °C to +85 °C
Storage temperature	−25 °C to +85 °C
IMU operating temperature	−40 °C to +105 °C
IMU storage temperature	−65 °C to +150 °C
IMU mechanical shock	2000 g, 0.5 ms ½ sine, any axis, unpowered

MECHANICAL	
Dimensions	22.5 × 25.0 × 7.1 mm
Weight	4 g

AT A GLANCE	
IMU	ADIS16507-2 / -3
Gyro	±500 / ±2000 °/s
Accel	±40 g
Interface	SPI
Supply	5 V
Dimensions	22.5 × 25 × 7.1 mm
Weight	4 g
Hardware rev	Rev 1

COMPLIANCE & SOURCING

› Designed and manufactured in the USA

ORDERING & RESOURCES

STORE arkelectron.com/product/ark-adis16507-2/
arkelectron.com/product/ark-adis16507/

DOCS docs.arkelectron.com/products/imu/ark-adis16507

CAD github.com/ARK-Electronics/ark-hardware

VARIANTS

	ARK ADIS16507-2	ARK ADIS16507-3
SKU	ARK-ADIS16507-2	ARK-ADIS16507-3
Gyro range	±500 °/s	±2000 °/s
Gyro in-run bias stability	2.7 °/h	7.5 / 8.1 / 4.9 °/h (X / Y / Z)
Accel in-run bias stability	12.7 / 12.7 / 13.7 µg	12.7 / 12.7 / 13.7 µg

Same board, connector and pinout. Bias stability is the ADIS16507 data sheet typical.

CONNECTOR PINOUT

PIN	SIGNAL	NOTES
1	VDD_5V	5 V supply
2	SPI_SCK	
3	SPI_MISO	
4	SPI_MOSI	
5	SPI_nCS	
6	NC	
7	SPI_nSYNC	
8	SPI_DRDY	Data ready
9	NC	
10	SPI_nRESET	Reset, 10 kΩ pull-up to 3.3 V
11	GND	

Connector: 11-pin JST-GH (SM11B-GHS-TB). Matches the ARK PAB SPI pinout (nCS2 / DRDY2 unused). Signals are 3.3 V.

NOTES

- Factory calibrated over −40 °C to +85 °C. Extended exposure outside −40 °C to +105 °C can degrade calibration accuracy.
- IMU environmental ratings are the ADIS16507 absolute maximum ratings.

ARK-DS-ARK-ADIS16507 | Version 1.0, 2026-10-29 | Compiled from the ARK ADIS16507 Carrier Production schematic Rev 1, the Production bill of materials, manufacturing documentation, and ARK Electronics published documentation. IMU specifications per the Analog Devices ADIS16507 data sheet Rev. A. Board temperature ratings per component data sheets. Specifications are subject to change without notice. © 2026 ARK Electronics LLC.